CH1101: Quantum Chemistry & Spectroscopy

22/09/2026

Lec 12

Particle in a sphere

H-like Atoms (H, He^+, Li^{2+} ....) with one electron

TISE in 3D $\quad \hat{H}\psi = E\psi$

Cartesian coordinate

$$\left[-\frac{\hbar^2}{2m}\nabla^2 + V(x,y,z)\right]\psi(x,y,z) = E\psi(x,y,z)$$

"del-squared" OR

∇^2: Laplacian operator

$$\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}$$

↓ 1. coordinate transformation
↓ 2. separation of variables

$$\psi_{n,l,m_l}(r,\theta,\phi) = \underbrace{R_{n,l}(r)}_{\text{Radial function}}\ \underbrace{Y_{l,m_l}(\theta,\phi)}_{\text{Angular function}}$$

Polar coordinate

n : 1, 2, 3 ..., n

l : 0, 1, 2, 3 .. (n-1)

m_l : 0, ±1, ±2, ... ±l

$0 \leq r < \infty$

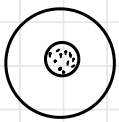
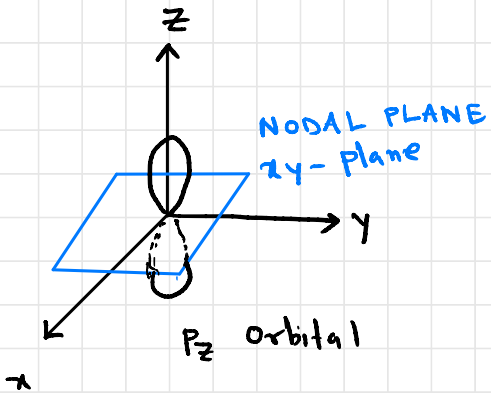
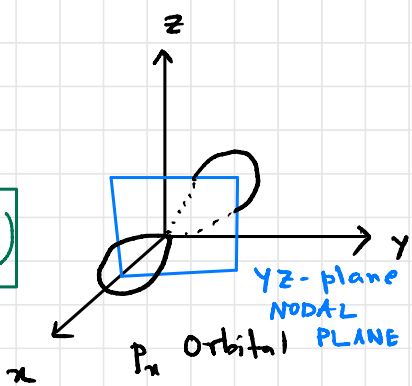
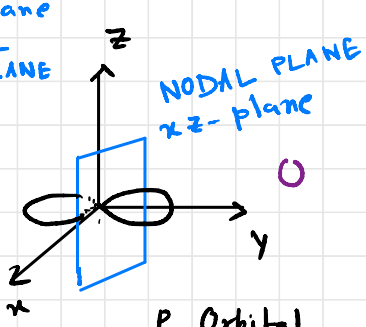
$0 \leq \theta \leq \pi$

$0 \leq \phi \leq 2\pi$

No. of Radial Nodes = $n - l - 1$

No. of Angular Nodes = l

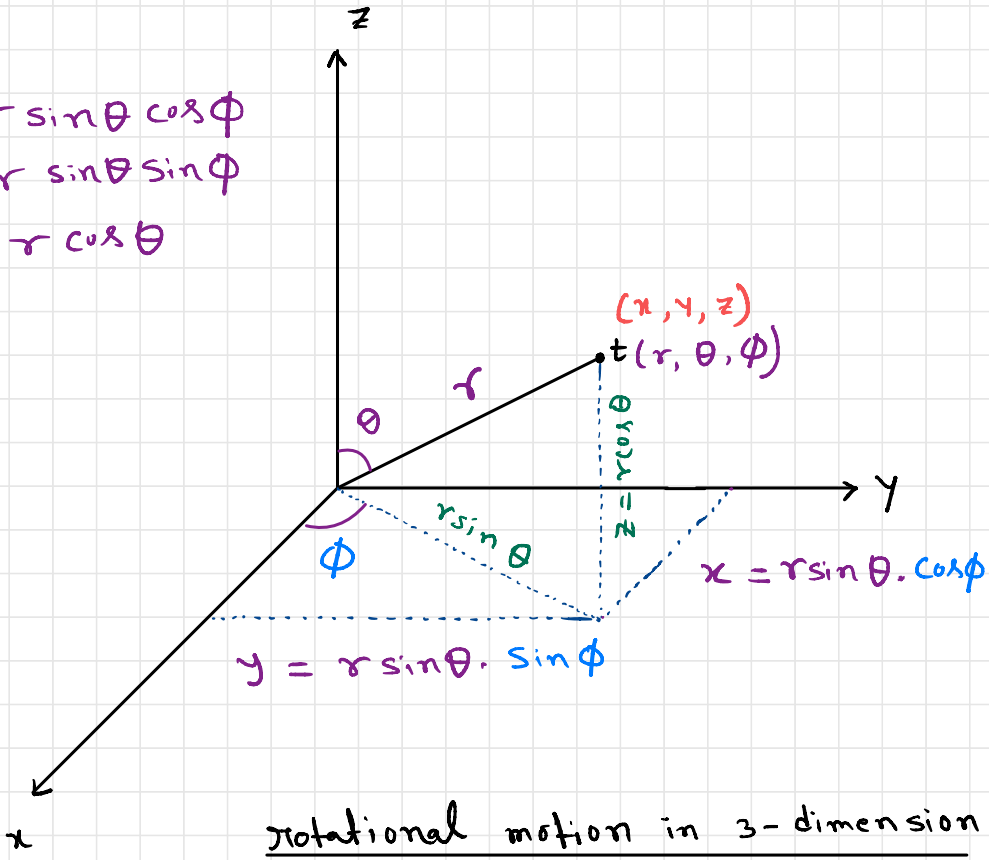
NODE : NO electron density found

Wavefunction	Atomic Orbitals	No. of Radial Nodes	No. of Angular Nodes
Ψ_{100}	 1s	0	0
Ψ_{200}	 2s	1	0
Ψ_{210}	 p_z orbital	0	1
$\Psi_{21\pm1}$ $p_x = \frac{1}{\sqrt{2}}(\Psi_+ + \Psi_-)$	 p_x Orbital	0	1
$\Psi_{21\pm1}$ $p_y = \frac{1}{\sqrt{2}}(\Psi_+ - \Psi_-)$	 p_y Orbital	0	1

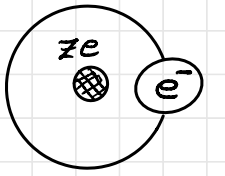
Reference

Spherical polar Coordinates

$$\begin{cases} x = r\sin\theta\cos\phi \\ y = r\sin\theta\sin\phi \\ z = r\cos\theta \end{cases}$$



Rotational motion in 3-dimension



Particle in a Sphere

